Math 120[1] (Introduction to Contemporary Mathematics)

Fall Semester 2026

Lecture: TuTh 1:00 – 2:15pm Online – Webex

Instructor: Ahmet Kaan Aydin
Office: Sondheim 305
Office hours: TuTh 5:00 – 7:00pm by appointment
Contact via email or Blackboard/Webex for virtual appointments.
e-mail: aaydin1@umbc.edu

Prerequisites: MATH 104, MATH 106, or a score of 3, 4, or 5 on the Math Placement Test.

Course Textbook: *Math in Society*, by David Lippman.

Content: Mathematical ideas through contemporary applications and real-world questions are explored.

1. **Rates and Growth:** percentages and decimals; absolute and relative change; linear growth; interpolation and extrapolation; regression and polynomial growth; exponential growth and logarithms; applications such as the Richter scale, Fibonacci growth, and population models; tax rates and interest; logistic growth and chaos.

2. **Probability and Statistics:** basic probability and probability rules; diagnostic-testing examples and the Monty Hall problem; expectation and mean; law of large numbers; normal distributions; mean and standard deviation; the 68–95–99.7 rule; sampling, confidence intervals, and opinion polls; correlation versus causation; proxies, machine learning, and big data.

3. **Mathematics in Society:** mathematical ideas related to elections, gerrymandering, voting paradoxes, and fair voting methods.

Testing and Grading: The usual grading system will be used in this course as follows:

Grade	Score (%)	Point Range
A	90% – 100%	450 - 500
B	80% – 89%	400 - 449
C	70% – 79%	350 - 399
D	60% – 69%	300 - 349
F	0% – 59%	0 - 299

Homework		50 points	(10%)
Classwork		50 points	(10%)
Midterm Exam		200 points	(40%)
Final Exam		200 points	(40%)
	TOTAL	500 points	(100%)

You're responsible for keeping track of assignment, classwork, and exam due dates.

Attendance and Behavior: Regular participation is important to success in this course because many topics will be developed through examples, discussion, and class activities. Attendance will be taken during scheduled synchronous meetings. If you miss a class, you remain responsible for the material, announcements, and deadlines posted in Blackboard.

Homework and Classwork: Coursework will follow the general structure of the lecture notes. **Reading (R)** assignments may be used to prepare for discussion, **Application (A)** assignments will focus on applying the mathematics developed in class, and **Classwork (C)** activities will be introduced regularly. Some classwork may be completed individually and some may be group-based. The exact number, format, and due dates of assignments may change as the semester develops and will be posted in Blackboard.

Midterm & Final Exam: There will be two exams.

- Midterm: **Tuesday, October 20, 2026 – In-person, room FA 303**
- Final: **TBA during the official UMBC final-exam period**

[1] Last updated: September 25, 2026

Academic Integrity: By enrolling in this course, each student assumes the responsibilities of an active participant in UMBC's scholarly community in which everyone's academic work and behavior are held to the highest standards of honesty. Cheating, fabrication, plagiarism, and helping others to commit these acts are all forms of academic dishonesty, and they are wrong. Academic misconduct could result in disciplinary action that may include but is not limited to suspension or dismissal. To read the full Student Academic Conduct Policy, consult the UMBC Student Handbook, the Faculty Handbook, or the UMBC Policies section of the UMBC Directory. `http://www.umbc.edu/provost/integrity/faculty.html`

Academic integrity is a core value at UMBC and I expect that value to be reflected in your coursework. You must cite any outside sources you use when required. You may not communicate with other people or use unauthorized resources while completing an exam. If you are unsure whether a resource or form of collaboration is permitted, ask before using it.

Accessibility and Disability Accommodations, Guidance and Resources: Accommodations for students with disabilities are provided for all students with a qualified disability under the Americans with Disabilities Act (ADA & ADAAA) and Section 504 of the Rehabilitation Act who request and are eligible for accommodations. The Office of Student Disability Services (SDS) is the UMBC department designated to coordinate accommodations that create equal access for students when barriers to participation exist in University courses, programs, or activities.

If you have a documented disability and need to request academic accommodations in your courses, please refer to the SDS website at `sds.umbc.edu` for registration information and office procedures.

SDS email: `disAbility@umbc.edu`
SDS phone: 410-455-2459

If you will be using SDS-approved accommodations in this class, please contact the instructor to discuss implementation of the accommodations.

Campus Safety: The following sections refer to campus policy on harassment, violence, and other issues. All policies and forms can be found at `ecr.umbc.edu`.

Sexual Assault, Sexual Harassment, and Gender Based Violence and Discrimination UMBC Policy in addition to federal and state law (to include Title IX) prohibits discrimination and harassment on the basis of sex, sexual orientation, and gender identity in University programs and activities. Any student who is impacted by sexual harassment, sexual assault, domestic violence, dating violence, stalking, sexual exploitation, gender discrimination, pregnancy discrimination, gender-based harassment, or related retaliation should contact the University's Title IX Coordinator to make a report and/or access support and resources. The Title IX Coordinator can be reached at `titleixcoordinator@umbc.edu` or 410-455-1717.

You can access support and resources even if you do not want to take any further action. You will not be forced to file a formal complaint or police report. Please be aware that the University may take action on its own if essential to protect the safety of the community.

If you are interested in making a report, please use the Online Reporting/Referral Form at `https://umbc-advocate.sympl` Please note that, if you report anonymously, the University's ability to respond will be limited.

Notice that Faculty and Teaching Assistants are Responsible Employees with Mandatory Reporting Obligations

All faculty members and teaching assistants are considered Responsible Employees, per UMBC's Policy on Sexual Misconduct, Sexual Harassment, and Gender Discrimination. Faculty and teaching assistants therefore are required to report all known information regarding alleged conduct that may be a violation of the Policy to the Title IX Coordinator, even if a student discloses an experience that occurred before attending UMBC and/or an incident that only involves people not affiliated with UMBC. Reports are required regardless of the amount of detail provided and even in instances where support has already been offered or received.

While faculty members want to encourage you to share information related to your life experiences through discussion and written work, students should understand that faculty are required to report past and present sexual harassment, sexual assault, domestic and dating violence, stalking, and gender discrimination that is shared with them to the Title IX Coordinator so that the University can inform students of their rights, resources, and support. While you are encouraged to do so, you are not obligated to respond to outreach conducted as a result of a report to the Title IX Coordinator.

If you need to speak with someone in confidence, who does not have an obligation to report to the Title IX Coordinator, UMBC has a number of Confidential Resources available to support you:

Retriever Integrated Health (Main Campus): `health.umbc.edu` 410-455-2472.

Center for Counseling and Well-Being (Shady Grove Campus): 301-738-6273.

Pastoral Counseling via The Gathering Space for Spiritual Well-Being: 410-455-6795; i3b@umbc.edu.

Other Resources

Women's Center (open to students of all genders): `https://womenscenter.umbc.edu/` 410-455-2714; womenscenter@umbc.edu.

Shady Grove Student Resources: `https://ecr.umbc.edu/shady-grove-title-ix-resources/`
Maryland Resources: `https://ecr.umbc.edu/maryland-resources/`
National Resources: `https://ecr.umbc.edu/national-resources/`

Child Abuse and Neglect

Please note that Maryland law and UMBC policy require that faculty report all disclosures or suspicions of child abuse or neglect to the Department of Social Services and/or the police even if the person who experienced the abuse or neglect is now over 18.

Pregnant and Parenting Students

UMBC's Policy on Sexual Misconduct, Sexual Harassment and Gender Discrimination expressly prohibits all forms of discrimination and harassment on the basis of sex, including pregnancy. Resources for pregnant, parenting and breastfeeding students are available through the University's Office of Equity and Civil Rights. Pregnant and parenting students are encouraged to contact the Title IX Coordinator to discuss plans and ensure ongoing access to their academic program with respect to a leave of absence, returning following leave, or any other accommodation that may be needed related to pregnancy, childbirth, adoption, breastfeeding, and/or the early months of parenting.

In addition, students who are pregnant and have an impairment related to their pregnancy that qualifies as disability under the ADA may be entitled to accommodations through the Office of Student Disability Services.

Religious Observances & Accommodations

UMBC Policy provides that students should not be penalized because of observances of their religious beliefs, and that students shall be given an opportunity, whenever feasible, to make up within a reasonable time any academic assignment that is missed due to individual participation in religious observances. It is the responsibility of the student to inform the instructor of any intended absences or requested modifications for religious observances in advance, and as early as possible. For questions or guidance regarding religious observances and accommodations, please contact the Office of Equity and Civil Rights at `ecr@umbc.edu`.

Hate, Bias, Discrimination and Harassment

UMBC values safety, cultural and ethnic diversity, social responsibility, lifelong learning, equity, and civic engagement.

Consistent with these principles, UMBC Policy prohibits discrimination and harassment in its educational programs and activities or with respect to employment terms and conditions based on race, creed, color, religion, sex, gender, pregnancy, ancestry, age, gender identity or expression, national origin, veterans status, marital status, sexual orientation, physical or mental disability, or genetic information.

Students (and faculty and staff) who experience discrimination, harassment, hate, or bias based upon a protected status or who have such matters reported to them should use the online reporting/referral form (`https://umbc-advocate.symplicity.com/titleix_report/index.php/pid954154?`) to report discrimination, hate, or bias incidents. You may report incidents that happen to you anonymously. Please note that, if you report anonymously, the University's ability to respond may be limited.

Class website: Course information and content (including this syllabus, assignments, readings, lecture materials, practice material, and announcements) will be available at the Blackboard website: `http://my.umbc.edu/go/black`